

What the data shows about the Upper East Side's Legionnaires' cooling towers

This summer, 76 buildings in Manhattan were ordered to scrub their rooftop cooling towers after tests found *Legionella* bacteria. We compared those buildings against their neighbors using the city's public records. Here's what stands out — in plain language.

Upper East Side • Carnegie Hill & Yorkville ZIP 10028 • 10075 • 10128 Updated July 2026



First, the reassuring part

The NYC Health Department says it remains **safe to shower, drink tap water, and use air conditioners** in the affected ZIP codes. Rooftop cooling towers are part of a building's air-conditioning system — separate from the water you drink and bathe in. Legionnaires' disease does not spread from person to person.

In early July 2026, health investigators noticed a cluster of Legionnaires' disease — a serious, treatable form of pneumonia — in the Carnegie Hill and Yorkville sections of the Upper East Side. They sampled water from more than 180 rooftop cooling towers in the area. Where a screening test came back positive for *Legionella*, the building was ordered to clean and disinfect its tower immediately, as a precaution. That list has grown to **76 buildings**.

Cooling towers matter because they release a fine mist into the outdoor air. If *Legionella* is growing inside one, that mist can carry the bacteria to people nearby. It's why cooling towers have been the source of past New York outbreaks — and why the city regulates, registers, and tests them.

What we looked at

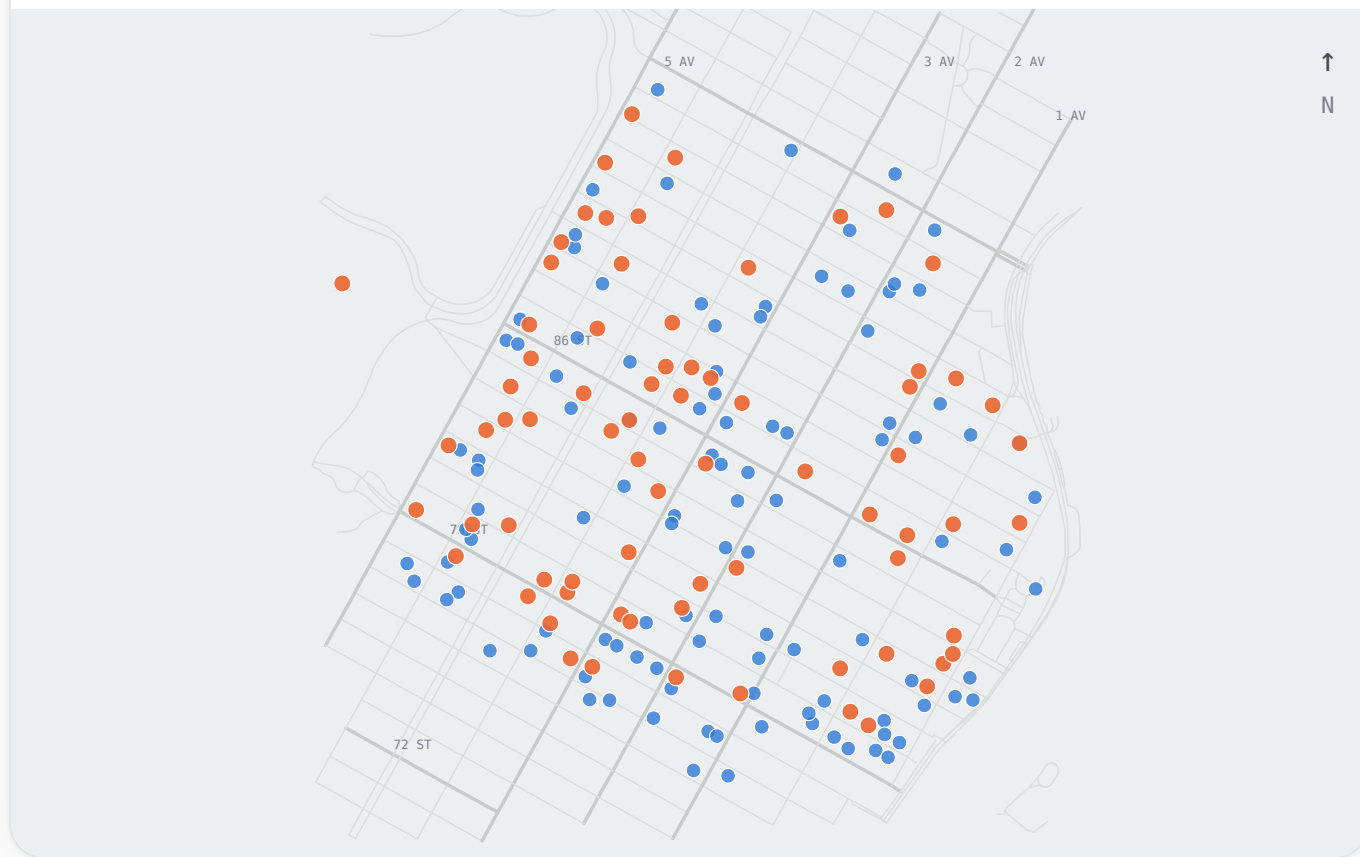
Every cooling tower in NYC has to be registered, tested for *Legionella* on a schedule, and inspected. All of that is public. We took the **76 buildings whose towers tested positive** and compared them against the **other registered cooling-tower buildings in the same three ZIP codes** — 183 buildings in all — asking a simple question: *do the positive buildings have anything different about them?*

We didn't stop at one or two things. We tested **about 40 factors** across six kinds of public records — testing and inspection history, violations of every type, building size and value, energy and water use, ownership and contractors, and location — using standard statistics and several study designs. The short version: almost nothing a regulator would flag explains which towers came back positive.

Positive and negative towers sit side by side

● Tested positive — ordered to clean (74)

● Registered tower in the same ZIPs, not on the DOH list (109)



Each dot is a building's cooling tower, at its real location. Hover (or tap) for details. Orange towers tested positive; blue did not. Notice they're mixed together — there is no single block or corner that stands out.

Four things the data says

01

The "40%" is about testing, not risk

You'll hear that ~40% of towers in these ZIP codes tested positive vs almost none elsewhere. That gap is mostly a **searchlight effect**: these ZIPs are the only place the city intensively screened, because that's where people fell ill. The ~0% elsewhere reflects *not being tested* — not being clean; *Legionella* can grow in cooling towers anywhere, especially in summer heat.

02

Not a story of neglect

Buildings with more violations, overdue tests, or fewer inspections were **not** more likely to test positive — even after matching buildings of similar size. The usual signs of a poorly-run building did not predict *Legionella*.

03

"Tests more" means diligent, not troubled

Positives leaned toward larger buildings that test their towers **more often** — and those buildings actually had *fewer* violations. So it looks like careful buildings catch more simply because they look more, not that they're worse-run.

04

No single hotspot

Positive and negative towers are **intermixed block by block** — no tight cluster, even by local hotspot tests. Consistent with bacteria turning up diffusely across the area rather than from one point.



One early, unconfirmed hint

The positive buildings tended to be slightly **taller than their neighbors** and a bit **closer to Central Park** — patterns that fit how tower mist drifts in open air. This is preliminary, could be a coincidence of small numbers, and would need public-health experts to confirm. It does *not* mean any particular building did something wrong.

Why the "40%" is misleading

The eye-catching contrast below is real, but it mostly measures **where the city looked**, not where the bacteria are. Investigators drew the testing boundary around where people got sick, then intensively sampled towers inside it. Towers elsewhere weren't part of this sweep — so they show ~0 "ordered to clean," not because they're clean, but because no one ran the same test on them.

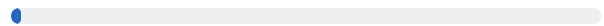
40%

of towers **tested here** came back positive



~0%

elsewhere — but those towers were **never screened** in this sweep



So the honest takeaway isn't "these blocks are uniquely dangerous" — it's "this is the one place that got a blitz test."

The factors that matter most — and aren't public

If maintenance records don't explain which towers bloomed, what does? Almost certainly the day-to-day condition *inside* each tower — and none of it is in public data:

- **Water temperature** in the tower basin — the single biggest driver of growth.

- **Chemical treatment** — the exact biocide schedule, and any lapse in it.
- **Stagnation** — a pump cycling off, an idle stretch, a dead leg in the pipes.
- **Biofilm and scale**, and the condition of the mist-catching parts.
- **The tower's make, model, and age** — some designs spray far more mist than others.
- **Lab culture results and patient locations** — kept private, and the key to naming a source.

Two buildings that look identical on every public record can differ completely on these. **That** is where the answer lives — a layer below anything the city publishes. Pinning down the exact source needs the Health Department's own lab and treatment data, not open records.

What this means for you

The cleaning orders are a **precaution**: a positive screening test doesn't guarantee live, infectious bacteria were present, so the city cleans first and confirms later. Our analysis found no evidence pointing at specific "negligent" buildings — so there's little to be gained from worrying about which address you live near. The sensible steps are the ordinary ones: follow the Health Department's guidance, and take symptoms seriously.

Know the signs

Legionnaires' disease usually appears **2-10 days** after exposure and is treatable with antibiotics, especially when caught early. Symptoms include:

- Fever, chills, cough
- Shortness of breath, chest discomfort
- Muscle aches, headache, sometimes diarrhea or confusion

Risk is higher for people **50 and older**, smokers, and those with chronic lung conditions or weakened immune systems. If you live or spent time in the area and have these symptoms, **see a doctor and mention the outbreak**. For questions, call **311**.

What this analysis can't tell you

- It's built on **which towers tested positive**, not on where sick people live — patient locations are private and never published.
- It can show patterns, but it **can't prove what caused** any individual infection.
- It's a snapshot of a fast-moving investigation; the official building list and case counts keep changing.
- With 74 positive buildings, the numbers are small — the findings are suggestive, not the final word.

How we did it. We matched the Health Department's public "clean & disinfect" list to NYC's Cooling Tower Registration and Inspection datasets on [NYC Open Data](#), then compared the positive buildings against other registered cooling-tower buildings in the same ZIP codes using standard statistics (logistic regression and a spatial clustering test). Building size and value come from the

city's PLUTO dataset. Street map from NYC's Street Centerline. A full technical writeup accompanies this piece.

Independent analysis — not an official record. This page is not affiliated with or endorsed by the NYC Health Department or any government agency. For official information, guidance, and the current building list, see nyc.gov/health or call 311. Nothing here is medical advice.